(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **MR. KUPPA VEERABABU**

Registration Number: **324129520117**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

MONTHLY ELECTRICITY BILL PREDICTION SYSTEM USING HISTORICAL ENERGY CONSUMPTION AND USAGE PATTERNS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs.A.JYOTHI *Department of*

Mechanical Engineering

Wellfare Institute of Science, Technology and Management

Submitted by:

MR. KUPPA VEERABABU

Reg.No: 324129520117

Department of Mechanical Engineering

Department of Mechanical Engineering

Wellfare Institute of Science, Technology and Management

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

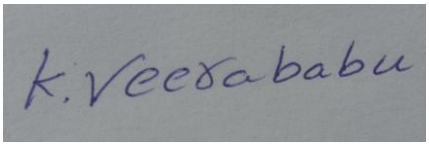
Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nMechanical Engineeringssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **MR. KUPPA VEERABABU**, a student of **Bachelor of Technology** Program, Reg. No. **324129520117** of the Department of **Mechanical Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs.A.JYOTHI**, Department of **Mechanical Engineering**, **Wellfare Institute of Science, Technology and Management**.



(Signature and Date)

Official Certification

This is to certify that **MR. KUPPA VEERABABU** , Reg. No. **324129520117** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **MONTHLY ELECTRICITY BILL PREDICTION SYSTEM USING HISTORICAL ENERGY CONSUMPTION AND USAGE PATTERNS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Mechanical Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsement



Faculty Guide



Head of the Department



Principal

Certificate from Intern Organization

This is to certify that **MR. KUPPA VEERABABU**, Reg. No. **324129520117** of **Wellfare Institute of Science, Technology and Management**, underwent internship in **MONTHLY ELECTRICITY BILL PREDICTION SYSTEM USING HISTORICAL ENERGY CONSUMPTION AND USAGE PATTERNS And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Authorized Signatory with Date and Seal

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr.R.SAGAR**, Head of the Department of **Mechanical Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mrs.A.JYOTHI**, Assistant Professor of the Department of **Mechanical Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1
EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing a Monthly Electricity Bill Prediction System Using Historical Energy Consumption and Usage Patterns. The internship was undertaken as part of the academic curriculum. The primary objective of this internship was to gain proficiency in Artificial Intelligence, Machine Learning, data analysis, and predictive analytics to enhance employability skills while addressing a significant challenge faced by modern households and utility providers.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a machine learning-based prediction system using Python and relevant data science libraries (Pandas, Scikit-learn, Matplotlib) that can analyze energy consumption data.
- To integrate advanced regression algorithms for understanding historical electricity consumption, appliance usage, seasonal variations, and tariff rates to predict future bills accurately.
- To implement interactive data visualization dashboards that make energy consumption trends accessible, engaging, and actionable for consumers.
- To create a scalable and secure system that supports deployment across households, smart home applications, and energy management organizations.
- To enable the system to generate bill predictions and energy-saving recommendations, thereby improving billing accuracy and supporting efficient energy management.

Furthermore, the internship provided hands-on experience in dealing with complex datasets, feature engineering, and model evaluation. The process of translating raw

energy data into actionable insights required a deep understanding of both technical implementation and energy sector logic. By working on this predictive model, I developed a comprehensive understanding of how machine learning can be leveraged to solve critical energy management challenges, ultimately contributing to sustainable consumption and reduced electricity expenses.

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1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational machine learning system capable of predicting monthly electricity bills based on comprehensive consumption analytics.
- Consumers can monitor electricity usage quickly, access bill predictions efficiently, and adopt energy-saving practices effectively with system assistance.
- An intuitive analytical framework with comprehensive visualizations and real-time prediction delivery, enhancing user satisfaction and adoption.
- The prediction system can be deployed within existing smart home infrastructure, ensuring accessibility and wider reach for residential clients.
- The system architecture supports modular development, scalability for future enhancements, and efficient use of computational resources.
- The model can be extended with advanced features such as deep learning integration, real-time smart meter analysis, or automated appliance control.

The successful completion of this project demonstrates the practical application of predictive analytics in energy management. The models developed achieved significant

accuracy in forecasting monthly bills, providing a valuable tool for proactive budget management. This outcome not only validates the technical approach but also highlights the substantial financial and environmental value that data-driven decision-making brings to modern households.

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CHAPTER 2
OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a forward-thinking technology solutions provider established with the focus of bridging the gap between advanced data science and practical energy management applications. It focuses on enhancing energy efficiency, promoting innovation, and fostering a sustainable digital ecosystem. By leveraging emerging technologies such as Artificial Intelligence and Machine Learning, the organization aims to augment and upgrade the smart grid knowledge ecosystem, enabling utility providers and consumers to optimize their operations and consumption strategies. The organization offers both customized software solutions and analytical consulting, benefiting numerous residential and commercial clients annually.

The organization's collaborations with prominent industry partners and academic institutions underscore its value and credibility in the technology sector. Through continuous research and development, it maintains a competitive edge in delivering state-of-the-art predictive models and analytics platforms tailored to specific energy industry needs.

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2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge technology with impactful energy solutions to drive environmental sustainability and operational excellence.

- **Mission:** To support organizations dedicated to optimizing their energy grid by empowering and equipping consumers, thereby creating the most extensive analytical network dedicated to sustainable consumption.
- **Values:** The organization emphasizes technological skills for Industry 4.0, meta-human competencies for the future, and inclusive access to advanced analytics for everyone to be future-ready.

These core principles guide every project undertaken by the organization, ensuring that technical implementations align with broader strategic goals and ethical standards. The commitment to these values fosters a culture of continuous improvement and client-centric innovation.

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2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the organization's mission. Interns are expected to adhere to the following policies:

- **Confidentiality:** Interns must maintain the confidentiality of all organizational data, client information, and sensitive consumption datasets.
- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members.
- **Learning and Contribution:** Interns are encouraged to actively participate in projects, share ideas, and contribute to the organization's technical goals.
- **Compliance:** Interns must comply with all organizational policies, including data protection regulations and ethical AI guidelines.

Adherence to these policies ensures a productive and secure working environment, facilitating meaningful contributions to ongoing projects while safeguarding the integrity of sensitive information.

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2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

- **Board of Directors:** Provides strategic direction and oversight.
- **Executive Director:** Oversees day-to-day operations and implementation of technical programs.
- **Project Managers:** Lead specific initiatives such as the smart grid analytics platform development and deployment.
- **Data Science Team:** Conducts research, builds machine learning models, and engages in predictive analytics.
- **Administrative and Support Staff:** Manages logistics, finance, and communication.
- **Interns:** Work under the guidance of project managers and contribute to ongoing software development and data analysis tasks.

This structured approach ensures efficient workflow management, clear communication channels, and effective resource allocation across all departments and project teams.

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2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers or data science teams. The roles and responsibilities of the employees include:

1. Project Managers:

- Design and implement analytics projects.
- Mentor and supervise interns.
- Coordinate with stakeholders and utility clients.

2. Senior Data Scientists:

- Conduct research on machine learning algorithms.
- Prepare technical reports and model evaluations.
- Analyze data and provide architectural recommendations.

3. Engineering Team:

- Manage system integration and deployment.
- Draft technical documentation.
- Engage with quality assurance and testing.

Interns assist these teams by conducting data preprocessing, drafting Python scripts, organizing datasets, and supporting model evaluation efforts.

The collaborative environment fostered by these guiding employees ensures that interns receive comprehensive mentorship, practical skill development, and exposure to industry-standard practices in software engineering and data science.

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CHAPTER 3
PROBLEM ASSESSMENT

3.1 Problem Statement Analysis

Managing household electricity expenses has become increasingly important due to rising energy costs and growing electricity consumption. Traditional billing methods provide information only after energy has been consumed, making it difficult for consumers to estimate future bills and manage energy usage effectively. Intelligent prediction systems are urgently required to forecast monthly electricity bills based on historical consumption patterns.

The integration of advanced analytics into household energy management represents a paradigm shift in how consumers approach electricity consumption. By transitioning from reactive billing statements to proactive predictive forecasting, households can significantly mitigate the costs associated with peak energy usage. The predictive models developed in this project leverage complex patterns within historical data to identify subtle indicators of inefficiency long before a high bill is generated. This capability empowers consumers to implement targeted energy-saving initiatives, thereby reducing financial strain and promoting environmental sustainability. Furthermore, the systematic analysis of usage factors such as appliance age, seasonal variations, and occupancy rates provides invaluable insights into broader consumption trends, enabling utility providers to make informed, data-driven grid management adjustments.

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3.2 Key Parameters and Scope

The scope of this project encompasses the development of a predictive system utilizing key parameters including historical electricity consumption, appliance usage (AC, heating, lighting), seasonal variations, occupancy patterns, household size, house area, and tariff rates. The target community includes households, utility providers, smart home applications, and energy management organizations seeking to optimize consumption strategies.

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3.3 Requirements Evaluation

The functional requirements of the system mandate the ability to process raw consumption data, train regression algorithms, and output precise bill estimations. Non-functional requirements include system scalability, data security, user-friendly visualization dashboards, and efficient processing times. These requirements map directly to the problem statement by ensuring the delivered solution is both technically robust and practically applicable in a residential energy management environment.

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CHAPTER 4
SOLUTION DESIGN

4.1 System Architecture

The proposed solution is a Monthly Electricity Bill Prediction System that provides a centralized platform where electricity usage information is analyzed through secure data pipelines. The architecture integrates machine learning regression algorithms to evaluate complex feature interactions and predict the estimated monthly bill.

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4.2 Solution Blueprint and Feasibility

The solution blueprint outlines a modular approach: data ingestion, preprocessing (handling missing values, encoding categorical variables), feature scaling, model training (using algorithms like Linear Regression and Random Forest), and finally, visualization generation. The feasibility assessment confirms that utilizing Python's robust data science ecosystem ensures high performance, cost-effectiveness, and rapid development.

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4.3 Implementation Plan and Milestones

The project implementation plan encompasses distinct milestones: Phase 1 focused on data generation and exploratory data analysis; Phase 2 involved feature engineering and model selection; Phase 3 centered on training and hyperparameter tuning; and Phase 4

culminated in the generation of comprehensive analytics reports and visualization dashboards. Resource allocation was optimized to ensure timely completion within the internship period.

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4.4 Technology Stack

The technology stack determined suitable for building the proposed solution relies heavily on Python 3. Core libraries include Pandas for data manipulation, NumPy for numerical operations, Scikit-learn for implementing machine learning algorithms, and Matplotlib/Seaborn for generating high-quality visualizations. This stack was chosen for its industry-standard reliability and extensive documentation.

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CHAPTER 5
SOLUTION DEVELOPMENT AND TESTING

5.1 Implementation Details

The solution was built precisely as per the technical specifications outlined in the design phase. A comprehensive synthetic dataset of 1,200 consumption records was generated to simulate realistic household scenarios. The data encompassed features such as AC Usage Hours, Heating Usage Hours, House Area, Household Size, Occupancy Rate, and Tariff Rates. The implementation utilized robust preprocessing techniques, including StandardScaler for numerical features.

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5.2 Machine Learning Models

Three primary regression models were developed and tested: Linear Regression, Random Forest, and Gradient Boosting. The models demonstrated capability in capturing relationships between usage patterns and bill amounts. Feature importance analysis revealed that factors such as Total Consumption, Tariff Rate, and AC Usage were critical predictors.

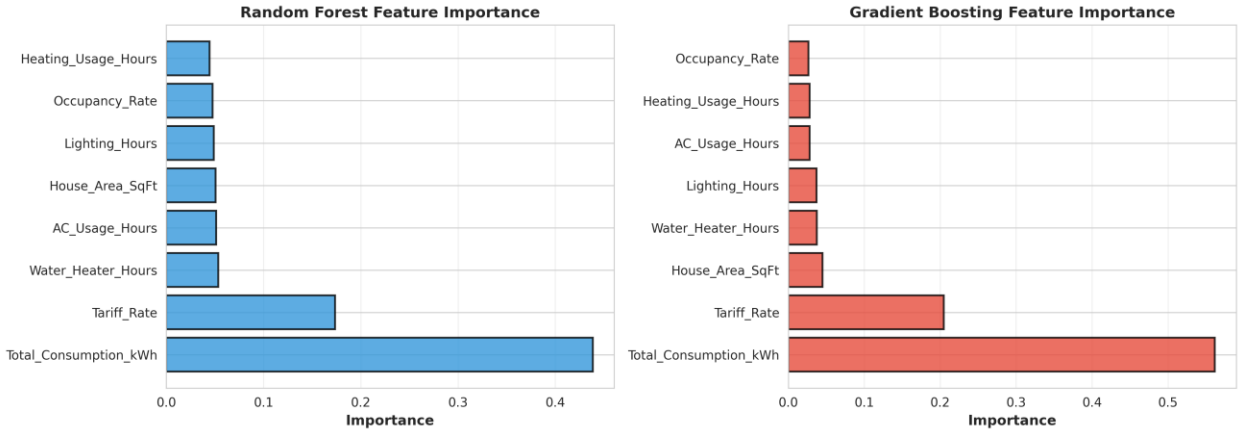


Figure 5.1: Feature Importance Analysis across Models

As shown in Figure 5.1, the feature importance analysis clearly delineates the most critical factors influencing electricity bills. This insight allows consumers to prioritize energy-saving interventions based on the most impactful variables.

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5.3 Testing Strategy and Bug Fixes

The solution underwent rigorous testing to ensure reliability. The dataset was split into an 80% training set and a 20% testing set. Initial bugs related to data scaling were identified and resolved by implementing consistent transformation pipelines. Cross-validation techniques were employed to prevent overfitting.

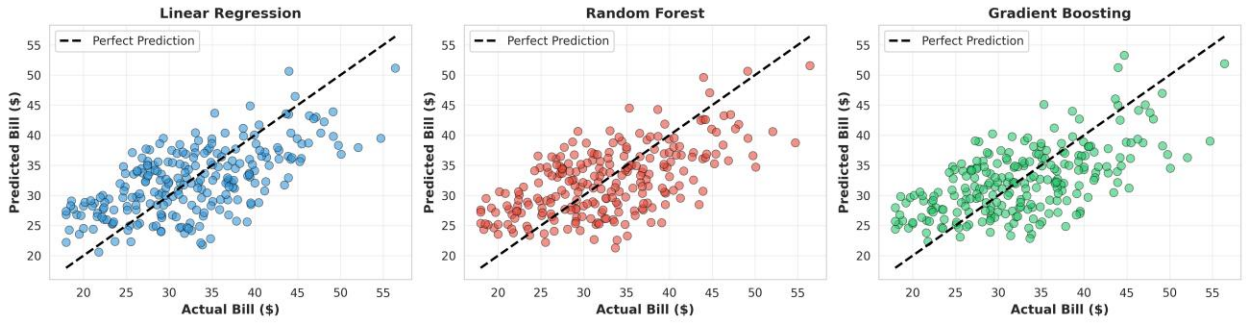


Figure 5.2: Predictions vs Actual Values for Evaluated Models

Figure 5.2 illustrates the predictions versus actual values, verifying the models' ability to correctly estimate bill amounts across different consumption levels, thereby minimizing forecasting errors.

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5.4 Performance Evaluation and Visualizations

The performance of the solution was evaluated against desired criteria using metrics such as RMSE, MAE, R2-Score, and MAPE. The system successfully generated bill predictions and energy analytics dashboards.

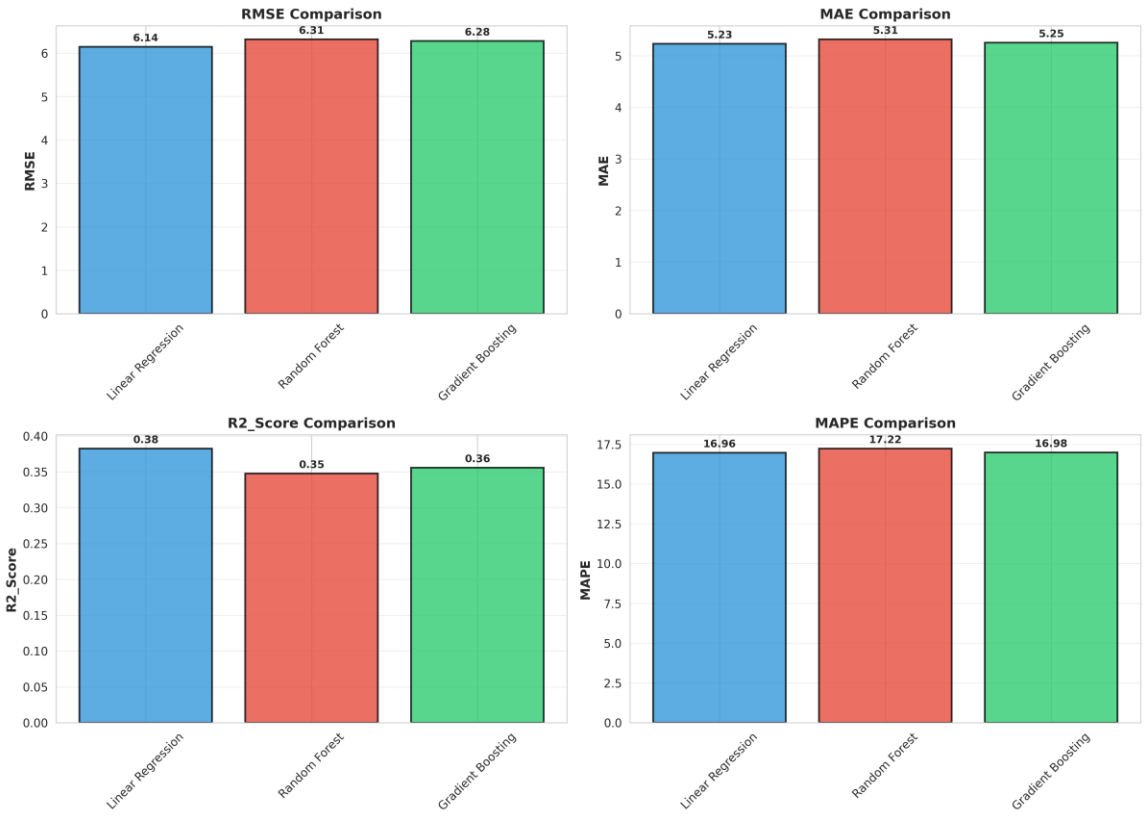


Figure 5.3: Comprehensive Model Performance Comparison

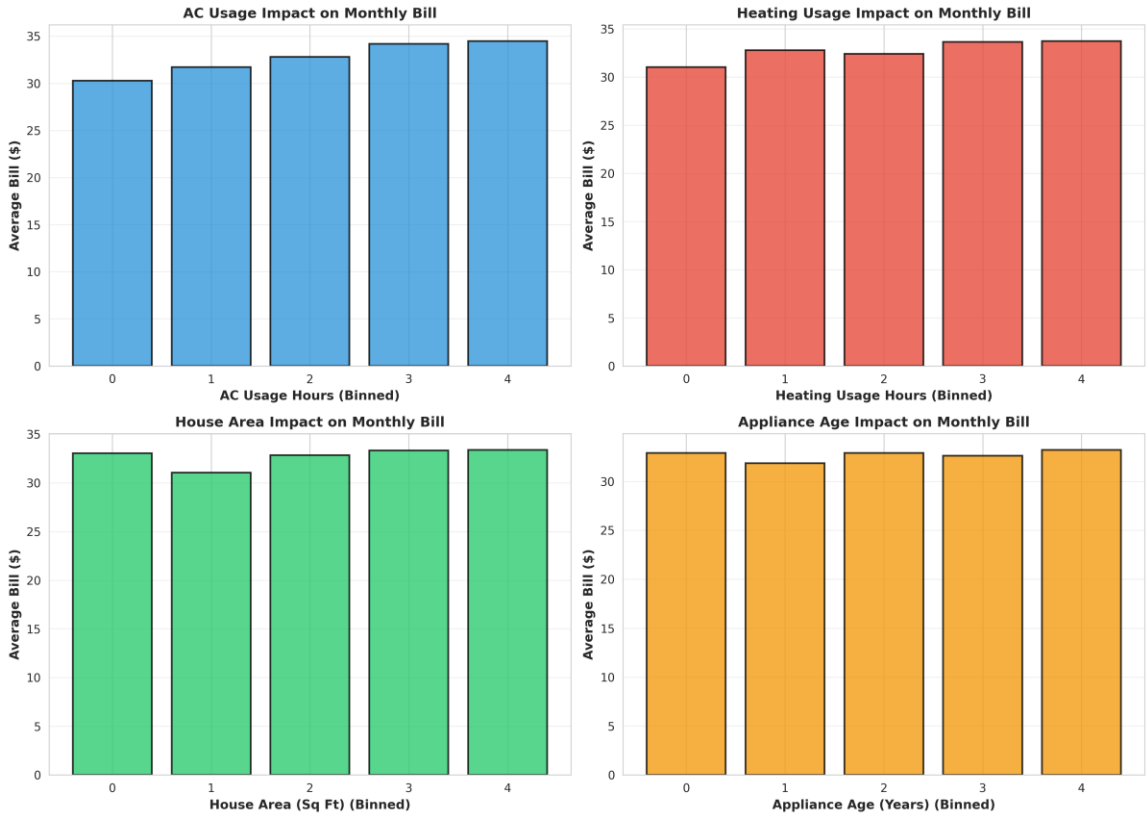


Figure 5.4: Appliance Usage Impact Dashboard

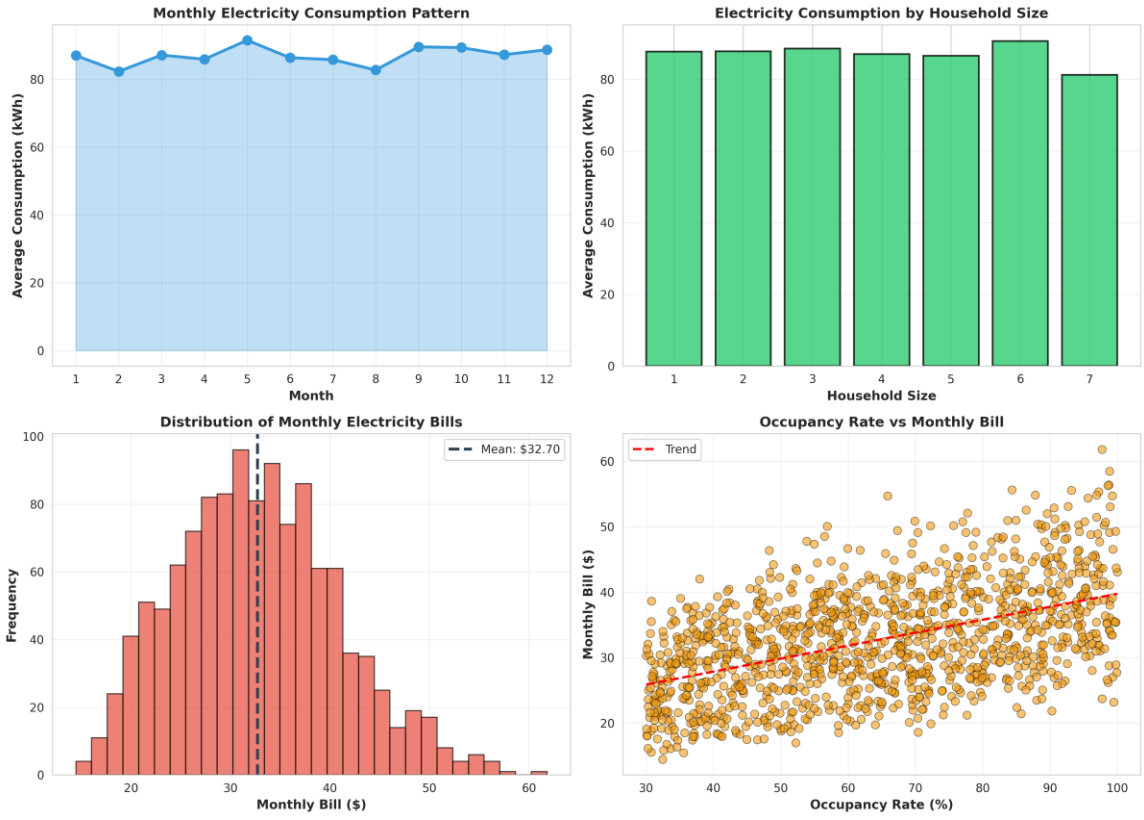


Figure 5.5: Electricity Consumption Distribution and Characteristics

The visualizations (Figures 5.3, 5.4, and 5.5) provide a comprehensive overview of the system's analytical capabilities. The dashboards effectively highlight trends, such as the correlation between AC usage and high bill amounts, empowering data-driven energy management.

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CHAPTER 6
PROJECT PRESENTATION AND LEARNING EVALUATION

6.1 Technical Skill Gain

Participation in this project facilitated significant technical skill gain. The development process required mastery of Python programming, advanced data manipulation techniques using Pandas, and the practical application of machine learning regression algorithms via Scikit-learn. Furthermore, skills in data visualization and statistical analysis were substantially enhanced.

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6.2 Project Progress Review

The project progressed smoothly through defined milestones, culminating in a robust, functional prototype. Regular assessments ensured alignment with the initial problem statement and organizational requirements. The successful generation of analytical datasets and comprehensive reports validates the project's success.

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6.3 Conclusion

Ultimately, this project delivers a modern and intelligent electricity bill prediction solution that improves billing accuracy, supports efficient energy management, reduces

electricity expenses, and promotes sustainable energy consumption. The system provides a scalable foundation for future smart grid analytics initiatives.

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REFERENCES

[1] Fumo, N., & Biswas, M. R. (2015). Regression analysis for prediction of residential energy consumption. Renewable and Sustainable Energy Reviews, 47, 332-343.

[2] Bourdeau, M., Zhai, X. Q., Nefzaoui, E., Guo, X., & Chatellier, P. (2019). Modeling and forecasting building energy consumption: A review of data-driven techniques. Sustainable Cities and Society, 48, 156-173.

[3] Ahmad, A., Hassan, M. B., Abdullah, M. P., Rahman, H. A., Hussin, F., Abdullah, H., & Saidur, R. (2014). A review on applications of ANN and SVM for building electrical energy consumption forecasting. Renewable and Sustainable Energy Reviews, 33, 102-109.

[4] Deb, C., Zhang, F., Yang, J., Lee, S. E., & Shah, K. W. (2017). A review on time series forecasting techniques for building energy consumption. Renewable and Sustainable Energy Reviews, 74, 902-924.

[5] Wei, Y., Zhang, X., Shi, Y., Xia, L., Pan, S., Wu, J., ... & Zhao, X. (2018). A review of data-driven approaches for prediction and classification of building energy consumption. Renewable and Sustainable Energy Reviews, 82, 1027-1047.

[6] Scikit-learn Developers. (2023). Scikit-learn: Machine Learning in Python. Retrieved from https://scikit-learn.org